

Nuforc Sightings Analysis

Savannah

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File Setup:

These are all the libraries needed for this analysis.

```
library(readr)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ purrr      1.0.4
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.5.2      ✓ tibble     3.3.0
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(stopwords)
library(RColorBrewer)
library(wordcloud)
library(tidytext)
```

Task 1: Building the Shape Table

Filtering and cleaning the data:

First, I will read in the data:

```
raw_df<- read_csv("nuforc_sightings.csv")
```

```
## Rows: 156711 Columns: 11
## — Column specification —————
## Delimiter: ","
## chr (10): link_url, occurred, city, state, country, shape, summary, reported...
## dbl (1): id
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
state_map <- read_csv("states.csv")
```

```
## New names:
## Rows: 50 Columns: 3
## — Column specification —————
## Delimiter: "," chr
## (2): state, name dbl (1): ...1
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## • `` -> `...1`
```

To filter the data, I used the states.csv for abbreviations of states. Additionally, some states mismatched to other countries so I corrected that in the filtering as well.

```
USA_df <- raw_df %>% mutate(state = tolower(state),
                           country = tolower(country),
                           shape = tolower(shape)) %>%
  filter(state %in% state_map$state | country == 'usa') %>%
  mutate(state = recode(state, "ohio" = "oh",
                        "new york" = "ny", "montana" = "mt",
                        "west virginia" = "wv", "wisconsin" = "wi")) %>%
  filter(state %in% state_map$state | state == "-" |
         state == "0" | state == "dc")
```

With this data, I standardized the shape column, removing all empty strings and turning them into “unknown”.

```
stand_shape <- USA_df %>%
  mutate(shape = ifelse(is.na(shape) | shape == "", "unknown", shape))
```

Creating the pivot table:

```
shape_by_state <- stand_shape %>% group_by(state, shape) %>%  
  summarise(count = n(), .groups = "drop")  
  
shape_state_pivot <- shape_by_state %>% pivot_wider(  
  names_from = shape, values_from = count, values_fill = 0)
```

There are 22 shapes in total excluding unknown and other. California/ CA saw the most circular shapes.

Task 2: PCA on the Shape Table

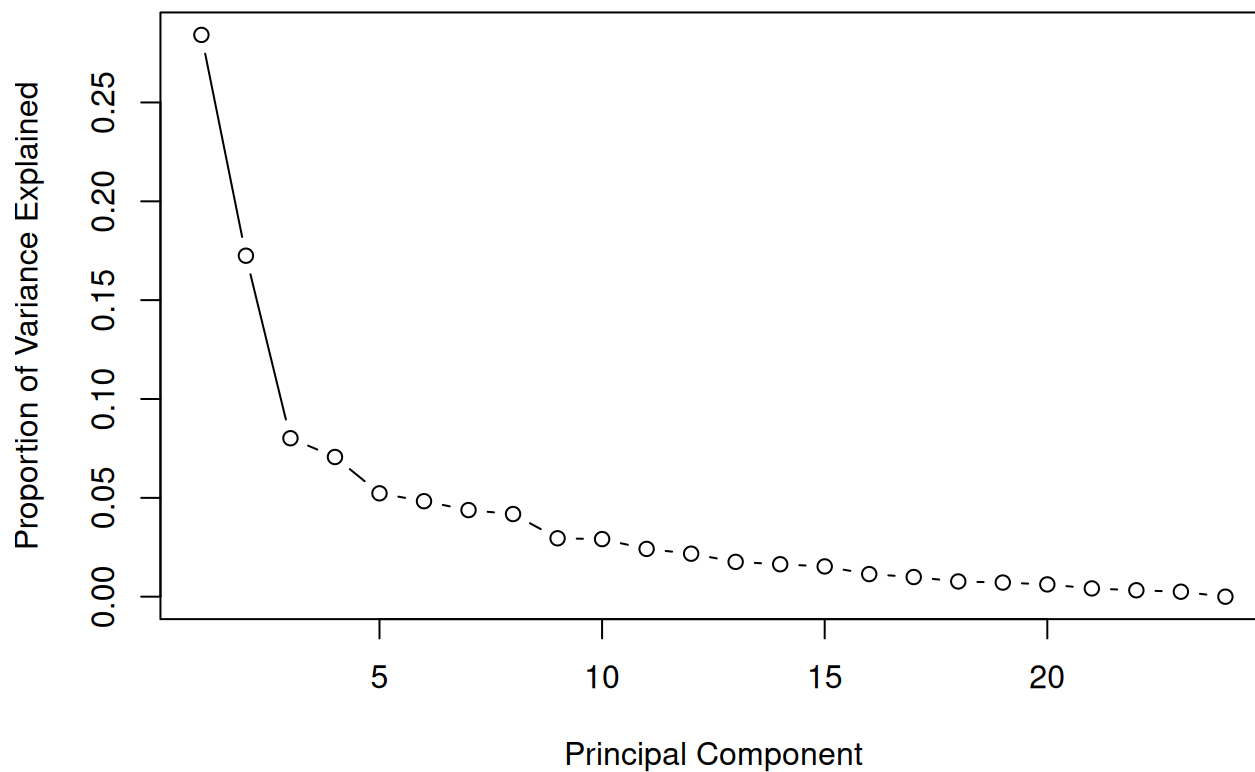
Row-normalizing the table and PCA:

```
#Row normalization  
shape_state_norm <- shape_state_pivot %>% rowwise() %>%  
  mutate(total = sum(c_across(-state))) %>%  
  mutate(across(-c(state, total), ~ .x / total)) %>%  
  select(-total) %>%  
  ungroup()  
  
# Performing PCA:  
pca_shape <- prcomp(shape_state_norm[, -1], center = TRUE, scale. = TRUE)
```

Proportion of variance explained and scree plot:

```
pve_shape <- (pca_shape$sdev)^2 / sum(pca_shape$sdev^2)  
  
plot(pve_shape, type = "b", xlab = "Principal Component",  
  ylab = "Proportion of Variance Explained", main = "Scree Plot")
```

Scree Plot

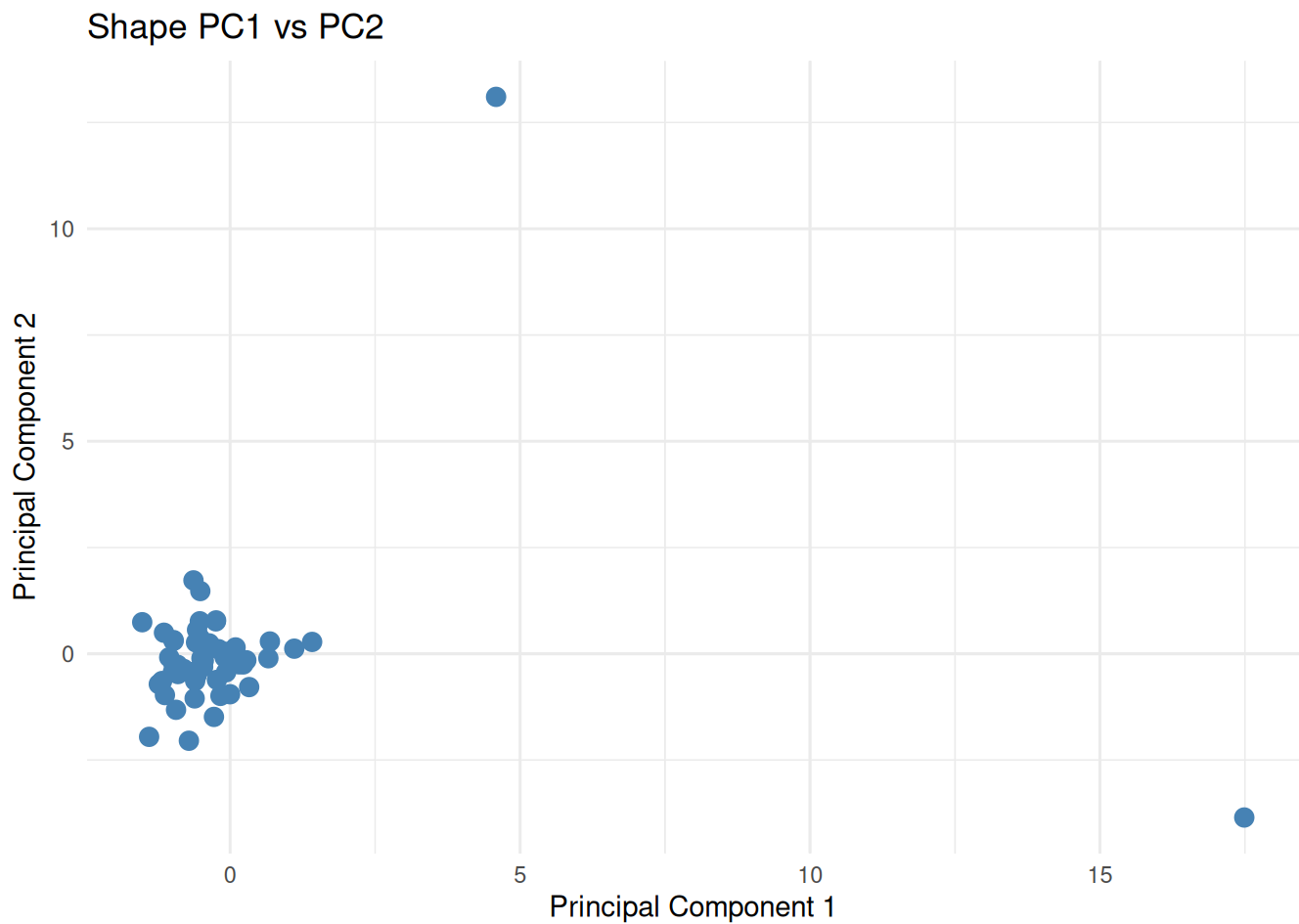


It does seem that the variety of UFO sightings can be boiled down to a few key patterns. Based on the scree plot, the proportion of variance explained by the first few principle components are enough to cover the majority of variance in the data.

Scatterplot:

```
pca_shape_df <- as.data.frame(pca_shape$x)

ggplot(pca_shape_df, aes(x = PC1, y = PC2)) +
  geom_point(color = "steelblue", size = 3) +
  labs(title = "Shape PC1 vs PC2",
       x = "Principal Component 1",
       y = "Principal Component 2") +
  theme_minimal()
```



Based on the scatterplot, the majority of states seem to be in one cluster. There are two states that exhibit outlier tendencies.

Examining the PCA rotation:

```
rotation_tbl <- as.data.frame(pca_shape$rotation) %>%  
  round(3) %>% .[,c(1,2)] %>%  
  rownames_to_column(var = "Variable")  
print(rotation_tbl)
```

##	Variable	PC1	PC2
## 1	changing	-0.199	0.277
## 2	cigar	-0.180	0.216
## 3	circle	0.351	-0.162
## 4	cross	-0.039	0.205
## 5	cube	0.080	0.430
## 6	cylinder	-0.076	0.305
## 7	disk	-0.278	-0.061
## 8	fireball	-0.252	-0.133
## 9	formation	-0.234	-0.051
## 10	light	-0.287	-0.113
## 11	orb	-0.200	0.188
## 12	other	-0.234	0.295
## 13	oval	-0.222	0.083
## 14	rectangle	-0.178	0.186
## 15	sphere	-0.273	-0.156
## 16	triangle	-0.219	-0.025
## 17	unknown	-0.019	0.409
## 18	chevron	-0.169	-0.134
## 19	cone	-0.147	-0.088
## 20	diamond	-0.229	-0.190
## 21	egg	-0.185	-0.176
## 22	flash	-0.192	-0.146
## 23	teardrop	-0.174	-0.122
## 24	star	-0.107	-0.093

It seems as though the shapes “light”, “sphere”, and “disk” affect PC1 the most. For PC2, the shapes “cylinder”, “cube”, and “unknown” contributed the most.

Task 3: Clean and Tokenize Summaries

To clean up the summaries:

```
Encoding(USA_df$summary) <- "UTF-8"
clean_summary <- USA_df %>%
  mutate(summary = tolower(summary)) %>%
  mutate(summary = iconv(summary, from = "UTF-8", to = "ASCII", sub = "")) %>%
  mutate(summary = gsub("\\s+", " ", trimws(summary)))
```

Making an array of words and removing punctuation:

```
word_array <- clean_summary %>% mutate(words = strsplit(summary, split = " "))
all_words <- unlist(word_array$words) %>% gsub("[[:punct:]]", "", .)
```

Frequency of each word:

```
freq_word <- table(all_words) %>% as.data.frame(.)  
colnames(freq_word) <- c("word", "frequency")  
freq_word <- freq_word[order(-freq_word$frequency), ]
```

Table and word cloud:

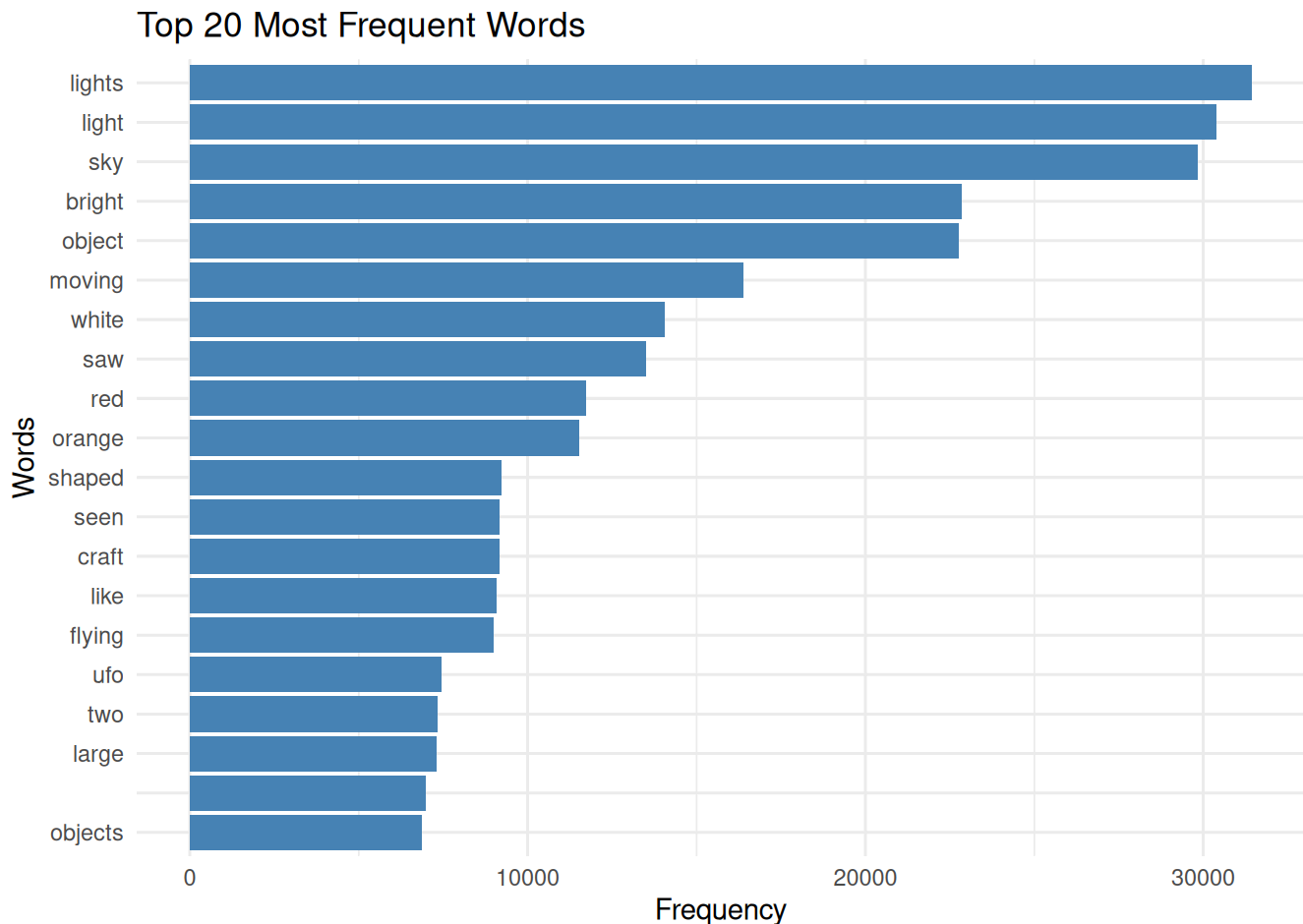
```
print(head(freq_word, 50))
```

##	word	frequency
## 43917	the	64752
## 6217	a	57344
## 24340	in	57309
## 7511	and	47415
## 32077	of	36315
## 26941	lights	31450
## 26778	light	30389
## 23911	i	29872
## 40188	sky	29839
## 44459	to	29086
## 47333	was	23749
## 10875	bright	22851
## 31634	object	22755
## 33028	over	20570
## 25156	it	20286
## 8501	at	18936
## 48484	with	17336
## 30041	moving	16402
## 32270	on	16289
## 47966	white	14063
## 43968	then	13775
## 38334	saw	13514
## 30368	my	13479
## 20704	from	12923
## 36449	red	11722
## 32446	orange	11539
## 43901	that	10015
## 46820	very	9796
## 39147	shaped	9217
## 38760	seen	9168
## 14449	craft	9163
## 27210	like	9073
## 20147	flying	9006
## 30970	no	7632
## 45695	ufo	7457
## 20315	for	7356
## 45608	two	7331
## 46240	up	7322
## 26279	large	7308
## 32863	out	7058
## 1		6983
## 31709	objects	6873
## 47539	we	6769
## 47697	were	6758
## 32282	one	6683
## 23475	hovering	6554
## 3521	3	6393

Histogram of cleaned words:

```
top_20 <- head(clean_word, 20)

ggplot(top_20, aes(x = reorder(word, frequency), y = frequency)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() + # Flips to horizontal bars
  labs(title = "Top 20 Most Frequent Words",
       x = "Words",
       y = "Frequency") +
  theme_minimal()
```



From the top frequent words after cleaning, we find the words “light”/“lights”/“bright”, “sky”, and “moving” as some of the top characterizations of UFO sightings

Task 4: Building the Keywords Table and PCA

Defining the dictionary of 100 words:

```
top_100 <- clean_word %>%
  filter(nchar(as.character(word)) >= 3) %>%
  slice_max(order_by = frequency, n = 100)
```

To make table of frequencies by state:

```
top100_df <- data.frame(word = top_100$word)
tokenized_df <- clean_summary %>%
  select(state, summary) %>%
  unnest_tokens(word, summary) %>%
  filter(word %in% top100_df$word)

word_state_pivot <- tokenized_df %>%
  count(state, word, sort = TRUE) %>%
  pivot_wider(names_from = word, values_from = n, values_fill = 0)

print(word_state_pivot)
```

```
## # A tibble: 53 × 101
##   state light   sky lights object bright moving white   red orange   saw like
##   <chr> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
## 1 ca      3899  3649  3558  3014  2780  1872  1846  1685  1597  1555  1472
## 2 fl      1786  1944  1781  1432  1368  1080   832   893  1169   871   729
## 3 wa      1831  1589  1501  1142  1388   917   857   733   728   729   547
## 4 az      1074  1111  1432   814   877   540   566   486   521   433   373
## 5 tx      1429  1385  1388  1144   952   755   709   491   440   667   509
## 6 ny      1320  1375  1308  1090  1002   763   657   607   617   680   447
## 7 pa      1146  1274  1247   896   931   674   616   606   650   522   399
## 8 il       946  1064  1132   737   680   533   434   595   495   448   293
## 9 oh       949  1019  1127   709   697   549   522   499   564   471   355
## 10 nc       875   823   993   645   657   528   414   419   483   380   276
## # i 43 more rows
## # i 89 more variables: shaped <int>, flying <int>, seen <int>, craft <int>,
## #   ufo <int>, two <int>, objects <int>, blue <int>, large <int>, nuforc <int>,
## #   note <int>, night <int>, green <int>, star <int>, hovering <int>,
## #   shape <int>, high <int>, one <int>, fast <int>, across <int>,
## #   triangle <int>, east <int>, north <int>, formation <int>, looked <int>,
## #   slowly <int>, west <int>, south <int>, three <int>, speed <int>, ...
```

Performing Task 2 for this table

Row-normalizing the table:

```
word_state_norm <- word_state_pivot %>% rowwise() %>%  
  mutate(total = sum(c_across(-state))) %>%  
  mutate(across(-c(state, total), ~ .x / total)) %>%  
  select(-total) %>%  
  ungroup()
```

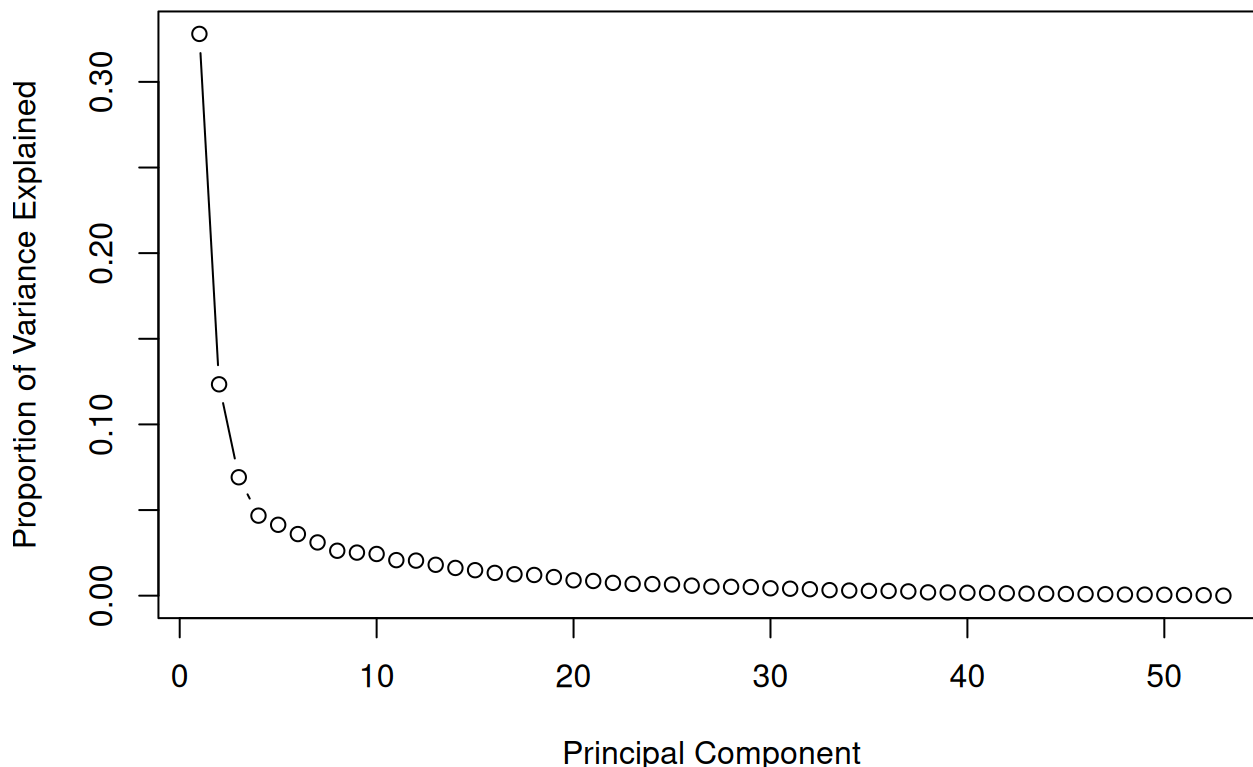
Performing PCA:

```
pca_word <- prcomp(word_state_norm[, -1], center = TRUE, scale. = TRUE)
```

Proportion of variance explained & scree plot:

```
pve_word <- (pca_word$sdev)^2 / sum(pca_word$sdev^2)  
plot(pve_word, type = "b", xlab = "Principal Component",  
     ylab = "Proportion of Variance Explained", main = "Scree Plot")
```

Scree Plot

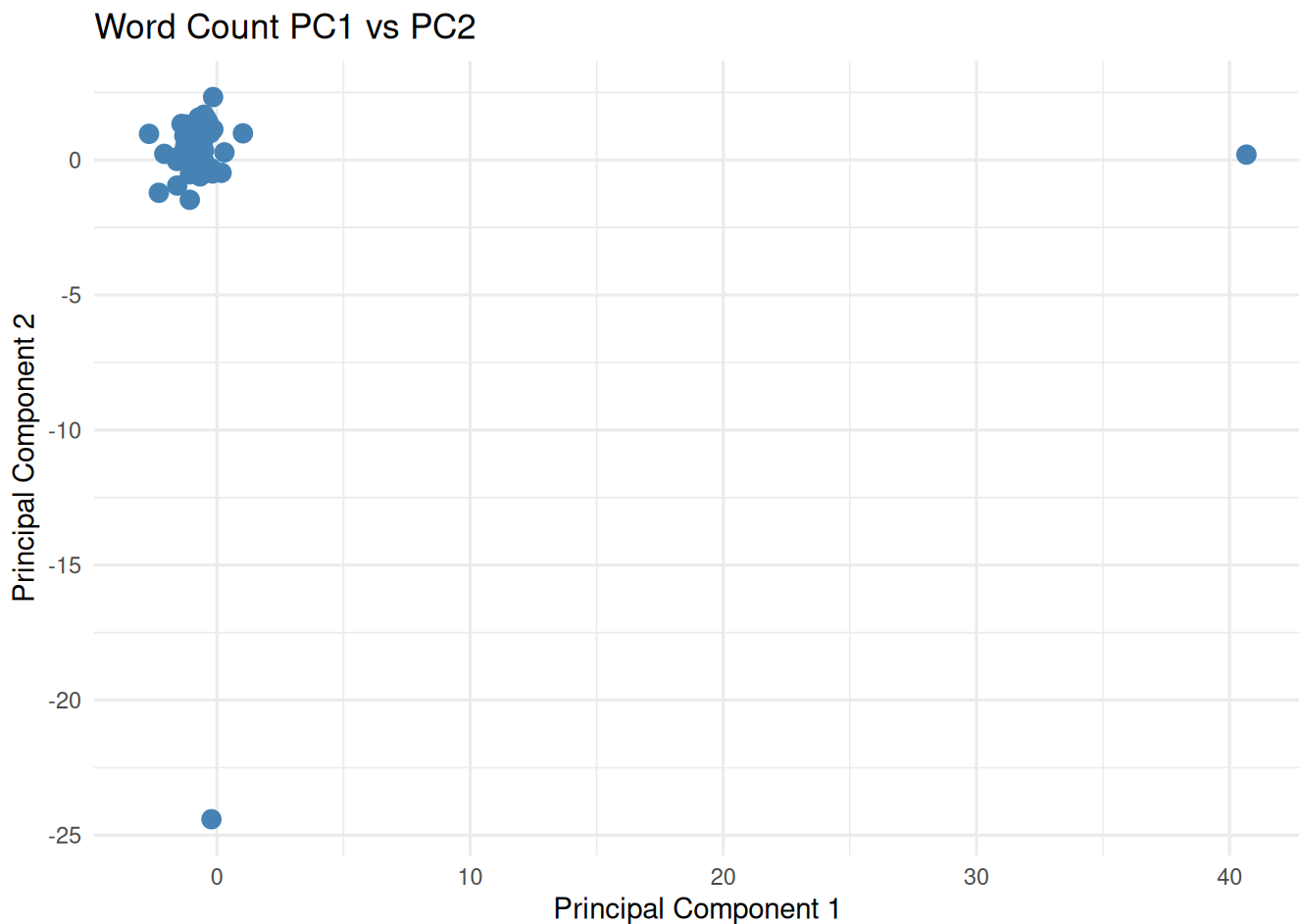


Similar to the shape by state data, this word count by state data also seems to be boiled down to a few patterns/ principle components. The majority of the variance is described by the first few principle components. The proportion of variance defined by latter PCs dwindle.

Scatterplot:

```
pca_word_df <- as.data.frame(pca_word$x)

ggplot(pca_word_df, aes(x = PC1, y = PC2)) +
  geom_point(color = "steelblue", size = 3) +
  labs(title = "Word Count PC1 vs PC2",
       x = "Principal Component 1",
       y = "Principal Component 2") +
  theme_minimal()
```



Similarly to the shape scatterplot, the majority of states seem to be in one cluster. There are again two states that seem to be outliers.

PCA rotation:

```
rotation_word_tbl <- as.data.frame(pca_word$rotation) %>%  
  round(3) %>% .[,c(1,2)] %>%  
  rownames_to_column(var = "Variable")  
print(rotation_word_tbl)
```

##	Variable	PC1	PC2
## 1	light	0.140	0.085
## 2	sky	0.139	0.119
## 3	lights	-0.129	0.030
## 4	object	-0.143	-0.013
## 5	bright	0.170	0.033
## 6	moving	-0.140	0.050
## 7	white	-0.117	0.145
## 8	red	-0.122	0.003
## 9	orange	-0.068	0.092
## 10	saw	0.144	-0.147
## 11	like	0.170	-0.049
## 12	shaped	-0.130	-0.046
## 13	flying	-0.118	0.030
## 14	seen	-0.102	-0.183
## 15	craft	-0.109	-0.124
## 16	ufo	-0.059	-0.237
## 17	two	-0.112	-0.078
## 18	objects	-0.105	0.070
## 19	blue	-0.089	0.100
## 20	large	-0.115	0.057
## 21	nuforc	-0.092	-0.094
## 22	note	-0.091	-0.093
## 23	night	-0.111	0.015
## 24	green	-0.095	-0.001
## 25	star	0.171	0.032
## 26	hovering	-0.117	0.047
## 27	shape	-0.117	-0.112
## 28	high	0.172	-0.018
## 29	one	-0.086	-0.182
## 30	fast	-0.113	0.012
## 31	across	0.170	0.036
## 32	triangle	-0.079	-0.172
## 33	east	-0.075	0.081
## 34	north	-0.067	0.113
## 35	formation	-0.091	0.118
## 36	looked	-0.104	-0.175
## 37	slowly	-0.103	0.057
## 38	west	-0.068	-0.006
## 39	south	-0.059	0.101
## 40	three	-0.095	0.165
## 41	speed	-0.099	0.080
## 42	glowing	-0.078	-0.144
## 43	ball	-0.089	0.015
## 44	flashing	-0.099	0.096
## 45	moved	-0.082	0.073
## 46	appeared	-0.099	0.091
## 47	around	-0.092	-0.165

## 48	disappeared	-0.111	0.043
## 49	sighting	-0.070	0.005
## 50	orb	-0.108	-0.048
## 51	fireball	-0.091	0.074
## 52	low	-0.082	0.138
## 53	round	-0.113	0.084
## 54	traveling	-0.098	0.020
## 55	strange	-0.086	0.140
## 56	triangular	-0.075	0.119
## 57	report	-0.045	-0.249
## 58	black	-0.076	-0.152
## 59	near	-0.074	0.125
## 60	sound	-0.098	-0.049
## 61	small	-0.087	0.133
## 62	just	-0.092	-0.129
## 63	sphere	-0.107	-0.027
## 64	circular	-0.094	0.056
## 65	looking	-0.084	-0.050
## 66	minutes	-0.084	0.150
## 67	observed	-0.063	-0.033
## 68	slow	-0.088	0.058
## 69	driving	-0.080	0.036
## 70	area	-0.074	0.117
## 71	noticed	-0.107	-0.041
## 72	anonymous	-0.082	-0.109
## 73	orbs	-0.074	-0.009
## 74	flew	-0.059	-0.190
## 75	away	-0.078	0.030
## 76	silent	-0.077	0.137
## 77	oval	-0.081	0.113
## 78	madar	-0.016	-0.001
## 79	aircraft	-0.079	-0.051
## 80	line	-0.094	0.083
## 81	see	-0.087	-0.181
## 82	stationary	-0.087	0.053
## 83	back	-0.086	-0.057
## 84	color	-0.101	0.028
## 85	direction	-0.099	0.029
## 86	side	-0.088	0.019
## 87	first	-0.050	-0.250
## 88	node	-0.016	-0.001
## 89	went	-0.099	-0.076
## 90	going	-0.081	-0.093
## 91	time	-0.077	0.020
## 92	straight	-0.085	-0.061
## 93	seconds	-0.078	0.028
## 94	blinking	0.172	0.014
## 95	stars	-0.069	0.004
## 96	house	-0.070	0.103

## 97	right	-0.094	-0.117
## 98	outside	-0.082	0.022
## 99	spotted	-0.100	0.008
## 100	home	-0.073	0.115

The words “bright”, “light”, “star”, “across”, “orb”, and “blinking” do suggest correlation with the shapes that were observed to be of primary drivers in the previous PCA. In general, the shapes that were most similarly described in this PCA are light”, “sphere”, “unknown” and “disk”. The shapes “rectangle” and “cube” are less accurately described by these top words.